



Numerical study of keyhole instability and porosity formation mechanism in laser welding of aluminum alloy and steel



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ABSTRACT

The instability of the keyhole and bubble being captured by solidification interface are the main factors that affect the formation of porosity in laser welding of both aluminum alloy and steel. In laser welding of aluminum alloy, the more violent fluid flow behind the keyhole, the larger depth to width ratio can cause the keyhole more unstable, and susceptible to collapse that can be described as three necessary steps. The much larger solidification rate and longer escaping distance make the bubble easily captured by the solidification interface, and easily combined. All of these reasons contribute to the formation of larger number and size of porosity in laser welding of aluminum alloy. The numerical and experimental results are in good agreement.

1. Introduction

Laser welding is extensively applied to the welding of structured materials, such as aluminum alloy and steel, in today's industrial manufacturing (Olubode et al., 2015). However, keyhole-induced porosity may occur during the welding, which would strongly weaken the joint strength (Yu et al., 2010; Lu et al., 2015). Owing to the great differences in the physical properties between aluminum alloy and steel, keyhole-induced porosity formation is usually expected to be different during laser welding of these two metals.

Matsunawa et al. (2003) used high speed X-radiography to observe the keyhole behavior during the laser welding of AA5083 aluminum alloy, and proposed that the dynamic pressure of metal vapor caused by intense evaporation of hump at the keyhole front wall, would generate dents at the rear wall of the keyhole, and accelerated the instability of the entire keyhole. The same phenomena can be observed in laser welding of steel (Matsunawa, 2001). Tsukamoto et al. (2003) used X-ray imaging system to observe the keyhole variation in 20 kW CO₂ laser welding of steel, and put forward the liquid column model to expound the porosity formation mechanism. Ribic (Ribic et al., 2009) suggested that the pressure equilibrium at the keyhole wall can affect the porosity formation. Courtois et al. (2016) and Meng et al. (2014) indicated the keyhole-induced porosity formation was closely related to keyhole collapse caused by keyhole instability. Based on experimental investigation, valuable information can be acquired to understand the

keyhole behavior and porosity formation. However, some useful information is hard to be obtained merely relying on experiments, such as the temperature distribution, the fluid flow field, which are closely related to the porosity formation and amalgamation.

Numerical simulation based on a comprehensive mathematical model can help better understand the process mechanism of laser welding. Many researches have been carried out to model the thermal and stress fields (Xia et al., 2014; Yilbas et al., 2010), and keyhole behavior (Volpp and Vollertsen, 2015; Mueller, 2012) in laser welding. However, few researches focused on modeling the keyhole-induced porosity formation. Zhou and Tsai (2007) developed a 2D mathematical models to systematically discuss the porosity formation in laser welding of 304 standard steels, and found that bubble was formed when the keyhole could not be filled by the molten metal. Zhao et al. (2011) simulated the keyhole phenomena and molten pool in laser welding of stainless steel 304L, and found that the keyhole collapse and shrinkage were the two factors that were responsible for keyhole-induced porosity. Cho et al. (2012) used the adiabatic bubble model to describe the bubble formation and motion during laser welding of low carbon steel. Lin et al. (2017) used numerical process simulations to investigate the effect of weld speed and laser inclination angle on porosity formation in laser welding of AA5182 aluminum alloy.

In this paper, a three-dimensional mathematical model considering the Fresnel absorption and multi-reflection of laser beam is developed to study the keyhole and bubble behaviors during the laser welding of

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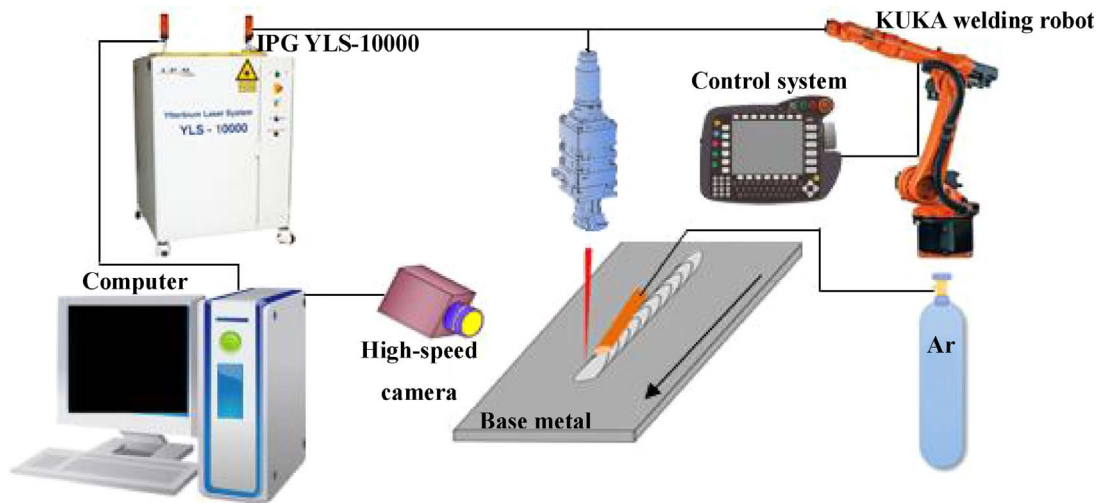


Fig. 1. Diagram of the experimental platform.

aluminum alloy and steel. The keyhole instability and porosity formation mechanism are analyzed. The differences in porosity formation between aluminum alloy and steel laser welding are also illuminated.

2. Experimental procedure

The experimental setup of 10 mm thick 5083 aluminum alloy and Q345 steel are used for the experiments is shown in Fig. 1, and the thermo-physical material properties of the base metal are summarized in Table 1. A fiber laser welding machine (IPG YLS-10000 with a maximum power of 10 kW) is used, and the wavelength of laser beam is 1.07 μm . Prior to welding, the plate is brushed by a stainless steel brush, and then cleaned by acetone. The optical fiber laser is focused on a spot with a diameter of 0.72 mm through a lens with a focal length of 400 mm. A high speed CCD camera is applied to observe the keyhole and weld pool behavior, and a low power laser-assisted light with a wavelength of 640 nm is used as backlight source to illuminate the welding zone to obtain clear images. The welding phenomenon is recorded at a frequency of 5000 frames/s.

The welding was performed at a welding speed of 2 m/min with a laser power of 6 kW. The defocused distance is 0 mm. In order to improve the welding quality, pure argon shielding gas with a flow rate of 20 L/min in an 8 mm diameter is blown into the weld pool at an angle of 45°. The X-ray radiography is carried out to analyze the porosity after welding. The longitude-section along the center line and the cross-section of the samples are observed by optical microscope of Zeiss Stemi2000 after etching in 10% HF.

3. Mathematical model and numerical simulation

Since laser welding involves an intricate multi-physical phenomenon, in present model, some assumptions and simplifications have

Table 1
Thermo-physical material properties used in simulation .

Nomenclature	Unit	aluminum alloy steel	
Density	Kg m^{-3}	2660	7830
Specific heat of the solid phase	$\text{J kg}^{-1} \text{K}^{-1}$	1050	670
Dynamic viscosity	Pa s	4.2×10^{-3}	5.9×10^{-3}
Thermal expansion coefficient	K^{-1}	1.5×10^{-4}	4.48×10^{-5}
Solidus temperature	K	847	1688
Liquidus temperature	K	933	1725
Latent heat of fusion	J kg^{-1}	3.87×10^5	2.77×10^5
Thermal conductivity	$\text{W m}^{-1} \text{K}^{-1}$	90	24.9

been considered as follows: (1) fluid flow is assumed to be Newtonian, laminar, and incompressible; (2) the weld shielding gas is neglected; (3) the laser heat source distribution of laser beam is assumed to be Gaussian distribution.

3.1. Governing equations

The differential equations governing the weld pool simulation of mass, momentum, and energy are applied to describe the heat transfer, mass transfer, and the melt flow. The Volume of Fluid (VOF) method is used to trace the free surface deformation of the weld pool (Li et al., 2014).

Mass conservation equation:

$$\nabla \cdot \vec{V} = 0 \quad (1)$$

where $\vec{V}$ is the velocity vector.

Momentum continuity equation:

$$\frac{\partial \vec{V}}{\partial t} + \vec{V} \cdot \nabla \vec{V} = \frac{1}{\rho} \nabla P + \mu \nabla^2 \vec{V} + K \vec{V} + F_b \quad (2)$$

where ρ is the fluid density, μ is kinematic viscosity, K is the drag coefficient, and F_b is body force.

Energy conservation equation:

$$\rho \frac{\partial h}{\partial t} + \nabla \cdot (\vec{V} h) = 0 \quad (3)$$

VOF equation:

$$\frac{\partial F}{\partial t} + \nabla \cdot (\vec{V} F) = 0 \quad (4)$$

where h is the enthalpy, κ is the thermal conductivity, and T is the temperature.

The energy-temperature relationship can be described by the solid-liquid phase transition model (Ronda and Siwek, 2011).

$$h = \begin{cases} \rho_s C_s T & (T \leq T_s) \\ h(T_s) + \Delta H_1 \frac{T - T_s}{T_1 - T_s} & (T_s < T \leq T_1) \\ h(T_1) + \rho_l C_l (T - T_1) & (T_1 < T) \end{cases} \quad (5)$$

where ρ_s , ρ_l are solid and liquid density respectively; T_s and T_1 are the solidus and liquidus temperature respectively; C_s and C_l are the specific heat of the solid and liquid phases respectively; ΔH_1 represent latent heat of fusion.

3.2. Driven forces

The main driven force of the weld pool is the recoil force, which can be expressed as follows:

$$P_r = 0.54P_0 \exp\left(L_v \frac{T - T_b}{RT_b}\right) \quad (6)$$

where P_r is the saturated pressure, P_0 is the atmospheric pressure, L_v is the latent heat of vaporization, T_b is the boiling temperature, and R is the universal gas constant.

The buoyancy force (body force) in the weld pool can be expressed as follow:

$$F_b = -g(T - T_0) \quad (7)$$

where g is the gravity, β is the coefficient of thermal expansion, and T_0 is the reference temperature.

The surface tension of the liquid metal can be expressed as a linear function of temperature:

$$\gamma = \gamma_0 + \frac{d\gamma}{dT}(T - T_0) \quad (8)$$

where γ_0 is the surface tension of pure metal at the melting point.

3.3. Laser welding model

The Gaussian distribution heat source model has been used due to its good versatility, which is expressed as follows (Lin et al., 2017):

$$q = Q \cdot \exp\left[-\left(\frac{r}{r_b}\right)^2\right] \quad (9)$$

where q denotes heat density function, Q is heat density value at the laser beam center, r is the radial coordinate and r_b is the radius of the laser beam at focus plane.

Multiple reflections of the laser beam can be calculated using the Fresnel absorption model, which is a function of the incident angle of the beam and a laser material-dependent coefficient. The reflectivity can be described as (Cho and Na, 2006):

$$\alpha(\varphi_N) = 1 - \frac{1}{2} \left(\frac{1 + (1 - \cos(\varphi_N))^2}{1 + (1 + \cos(\varphi_N))^2} + \frac{2 - 2\cos(\varphi_N) + 2(\cos(\varphi_N))^2}{2 + 2\cos(\varphi_N) + 2(\cos(\varphi_N))^2} \right) \quad (10)$$

where (φ_N) is the angle of reflection between the N th incident ray and the N th surface normal, and is a material-dependent quantity.

In this model, multiple reflections of the laser shown in Fig. 2 are considered inside the keyhole, and the calculation of the absorbed energy after the N th reflection is expressed as follows:

$$q_N = q'(N - 1)\alpha(\varphi_N) \quad (11)$$

3.4. Evaporation and bubble models

The material evaporated during welding, which resulted in energy loss on the surface of keyhole and weld pool:

$$q_{\text{vap}} = C_{\text{vap}}(T - T_v) \quad (12)$$

where q_{vap} is the energy loss, ρ is the density of the material, T_v is liquid us temperature.

The mass loss due to evaporation is expressed as follows:

$$M_{\text{net}} = R_{\text{accom}} \sqrt{\frac{M}{2RT_{\text{bdy}}}} (P_1^{\text{sat}} - P_v) \quad (13)$$

where M is the molecular weight of the vapor, R_{accom} is an evaporation coefficient, R the vapor gas constant, T_{bdy} is the average temperature of surface molten metal, P_v is vapor pressure. The superscript sat on P_1 can be calculated with Eq. (6).

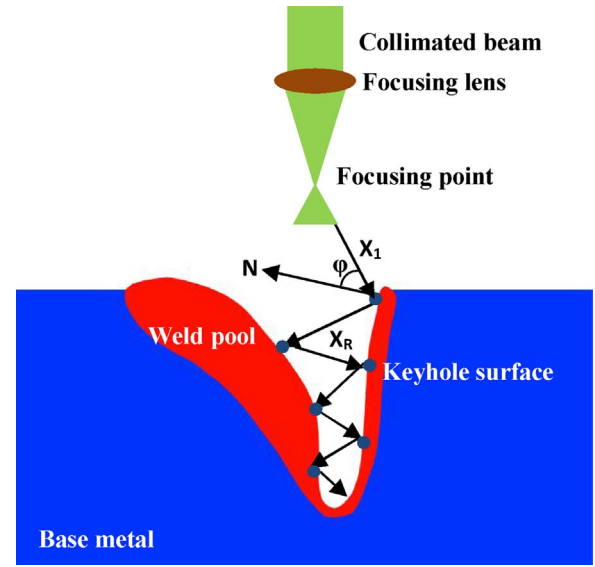


Fig. 2. Multiple reflections inside keyhole.

$$P_1^{\text{sat}} = P_{v1} \cdot \exp\left[\frac{-\left(\frac{1}{T} - \frac{1}{T_{v1}}\right)}{\left(\frac{C_p}{C_v} - 1\right) \frac{C_v^{\text{vap}}}{L_v}}\right] \quad (14)$$

where C_p and C_v are the specific heat at constant pressure and volume of the vapor respectively, L_v is latent heat of evaporation.

The bubble model is introduced to explain the transition of the bubbles into the porosity. The migration kinetics of the bubbles in the molten pool can be calculated by as follows:

$$\frac{\partial V_b}{\partial t} = -\frac{\nabla P}{\rho_b} + g + (V - V_b) + (V - V_b) \cdot (V - V_b) \cdot \frac{\beta}{b} \quad (15)$$

where P is the pressure in the bubble, V_b is the mean velocity of the bubble, ρ_b is the density of the gas in the bubble, V is the flow velocity of the fluid, V_b is the pressure in the molten pool. is the drag constant between the melt and bubbles, which can be given by Eq. (16):

$$\beta = \frac{3d}{4} \left(\frac{24}{\text{Re}} + \frac{6}{1 + \sqrt{\text{Re}}} + 0.4 \right) \quad (16)$$

where d represents the average diameter of the bubbles, Re is the Reynolds number of the fluid.

3.5. Boundary conditions

There are a variety of energy transfer modes on the keyhole surface, including heat dissipation by convection and radiation and heat loss due to heat transfer and surface evaporation.

On the top free surface of the keyhole, the energy balance can be expressed as follows (Zhang et al., 2014):

$$-k \frac{\partial T}{\partial n} = -q_{\text{laser}} + h_c(T - T_0) + \epsilon_r (T^4 - T_0^4) \quad (17)$$

where n is the vector normal to the local free surface, h_c is the convective heat transfer coefficient, σ is the Stefan-Boltzmann constant, and ϵ_r is the surface radiation emissivity.

For other surface:

$$k \frac{\partial T}{\partial n} = -h_c(T - T_0) - \epsilon_r (T^4 - T_0^4) \quad (18)$$

The pressure boundary of the top free surface is expressed as:

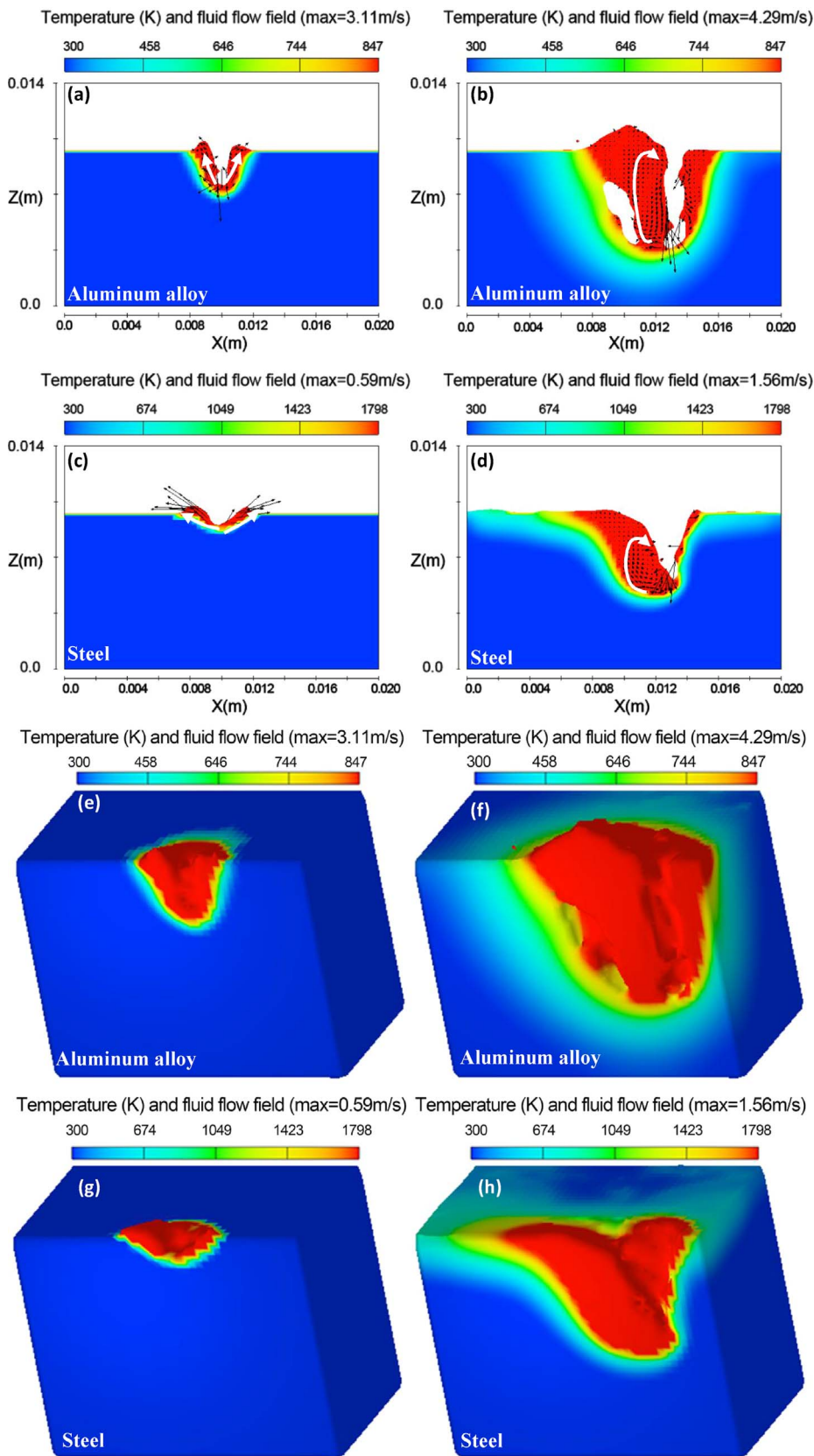


Fig. 3. Formation process of keyhole during laser welding at 6 kW (a) aluminum alloy at 0.005 s (b) aluminum alloy at 0.1 s (c) steel at 0.005 s (d) steel at 0.1 s (e)–(h) Three-dimensional views of weld pool, temperature field, and flow field correspond to longitudinal cross sectional (a)–(d), respectively.

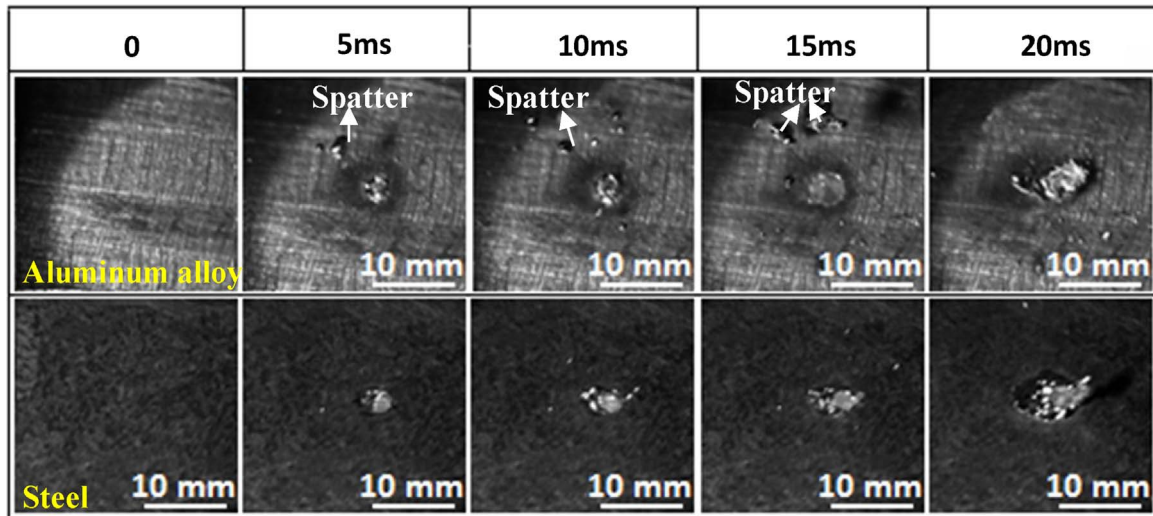


Fig. 4. High-speed images showing keyhole formation in laser welding of aluminum alloy and steel.

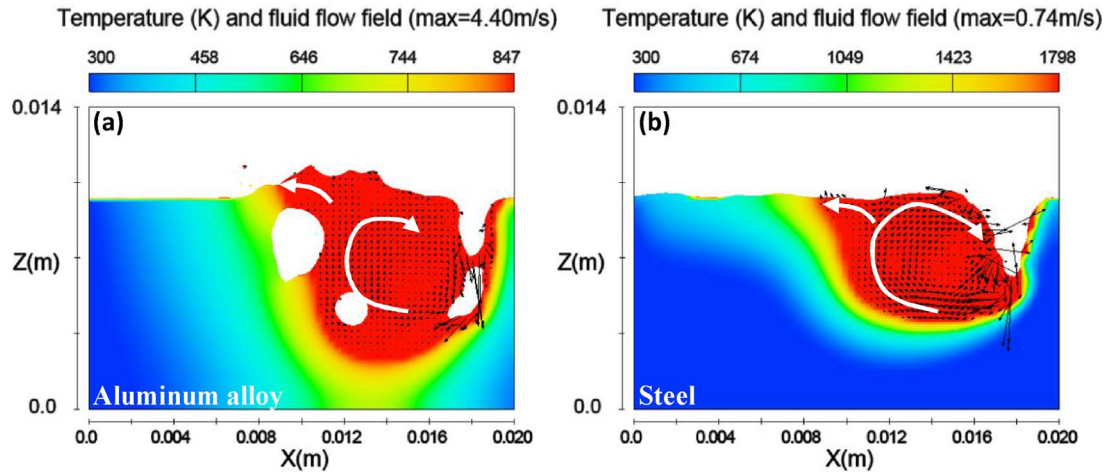


Fig. 5. The weld pool during laser welding at 0.25 s (a) aluminum alloy (b) steel.

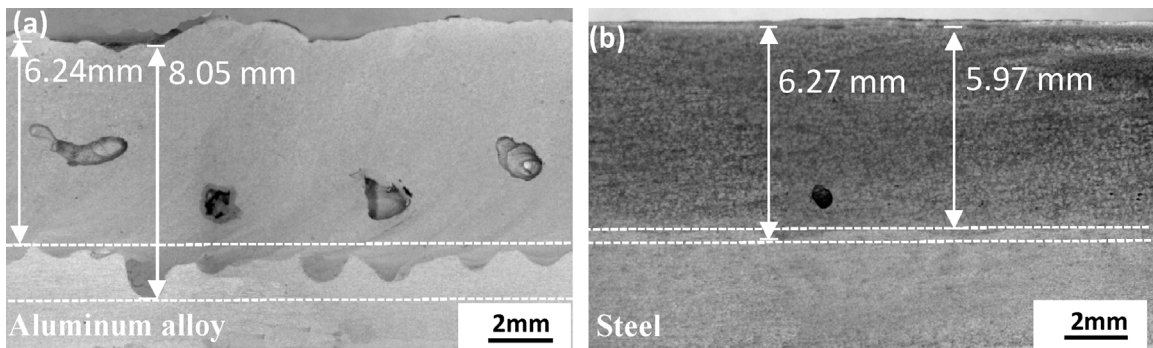


Fig. 6. the longitudinal sections of the weld beads: (a) aluminum alloy; (b) steel.

$$-p + 2\mu \frac{\partial \vec{V}}{\partial n} = p_r + \frac{\gamma}{R} \quad (19)$$

where p is hydrodynamic pressure, and $\vec{V}_n$ is normal velocity vector, $\vec{V}_n$ is the normal to the free surface, and R is the radius of surface curvature.

4. Results and discussion

4.1. Weld pool convection in laser welding

Fig. 3 shows the formation process of the keyhole in laser welding of aluminum alloy, in which the red regions denote the temperature is higher than the melting point of the material. In the initial stage, the workpiece is melt and evaporated due to the direct irradiation of laser beam, resulting in a small concave and pileups, as shown in Fig. 3(a). As the welding process continues, the temperature in the laser radius area

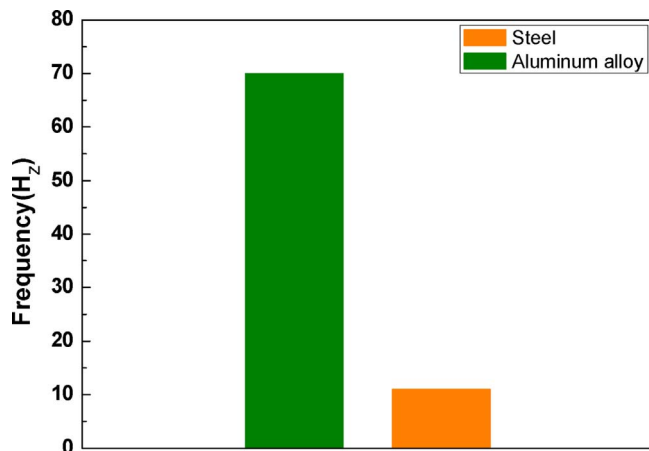


Fig. 7. Frequency of keyhole collapse in simulation.

is elevate rapidly, more and more molten metals are pushed out of the concave due to the reduced surface tension and increased recoil pressure to form a keyhole, as shown in Fig. 3(b). The width and depth of the weld pool are gradually outspreaded over the time, and ultimately fluctuate within limits. It can be seen from Fig. 3(c) and (d) that the flow patterns inside the weld pools are similar during laser welding of steel. Three-dimensional views of weld pool, temperature filed, and flow field correspond to longitudinal cross sectional Fig. 3(a)–(d) are shown in Fig. 3(e)–(h) to depict the weld pool convection in laser welding.

Fig. 4 shows images of keyhole formation captured by high speed camera in laser welding of aluminum alloy and steel. At $t = 5$ ms, small concaves are formed, a lot of molten metals are pushed out under the influence of recoil pressure caused by evaporation. At prolonged welding, the size of the concaves becomes larger, and keyholes are generated. The formation process of the keyhole during laser welding is similar for aluminum alloy and steel.

The laser welding pools at 0.25 s of aluminum alloy are shown in Fig. 5(a). On the front of the keyhole wall, the molten metal flows randomly. On the back keyhole wall, however, a clockwise vortex exists in the rear part of the weld pool, which is caused by the fact that the molten metal at the upper part of the weld pool flows downward along the keyhole wall, while the molten metal at the bottom part of the weld pool flows upward along the fusion line under the effect of recoil pressure. The clockwise vortex can assist the molten metal heated by the laser beam transport to the rear upper part of the weld pool. The

molten metal at the rear part of the top surface flows towards the weld pool edge due to the effect of surface tension gradient. The flow around the keyhole is more drastic than other parts of the weld pool. Although many physical properties differ between aluminum alloy and steel, the flow patterns inside the weld pools during laser welding are similar, as shown in Fig. 5(b). However, owing to the lower viscosity and surface tension and higher evaporation rate of 5083 aluminum alloy, the melt flow in laser welding of 5083 aluminum alloy is much stronger than that of steel.

4.2. Keyhole instability in laser welding

The longitudinal sections of the weld beads are shown in Fig. 6. It can be seen that both the minimum and maximum weld depths of aluminum alloy (6.44 mm and 8.05 mm, respectively) are higher than those of steel (5.97 mm and 6.27 mm, respectively). Besides, the weld depth is violently fluctuated in the welding direction.

Fig. 7 shows the frequency of keyhole collapse derived from numerical results. The frequency of keyhole collapse during laser welding in the aluminum alloy is much more than that in steel, i.e., 70 Hz vs. 11 Hz.

Fig. 8 shows images of keyhole and weld pool behaviors. During laser welding of aluminum alloy, the weld pool and the keyhole are fluctuated drastically, more spatters are formed around the keyhole.

According to Pang et al. (2011) study, the mechanism of keyhole stability is closely correlated with fluctuations of weld depth. Higher fluctuation of weld depth, more frequent collapse of keyhole, and larger number of spatters formed around the keyhole indicate that during aluminum alloy laser welding, the keyhole is more instable. The reasons are discussed as follow.

- (1) The melt flow in the weld pool has great influence on keyhole stability. A sequence of images of the keyhole collapse is shown in Fig. 9, from which it is readily observed that the formation of keyhole collapse in both aluminum alloy and steel laser welding pools are similar. As shown in Fig. 9(a) and (c), the molten metal behind the upper rear keyhole wall flows toward the front keyhole wall under the influence of strong clockwise flow. As a result, bulge is formed at the rear wall of the keyhole. During laser welding of aluminum alloy, due to the lower surface tension and viscosity of 5083 aluminum alloy, and higher evaporation rate caused by the volatile magnesium in the weld pool, the flow behind the upper rear keyhole wall is more intense, and thus bulge with larger size is easily formed at the upper rear keyhole wall. The bulge will

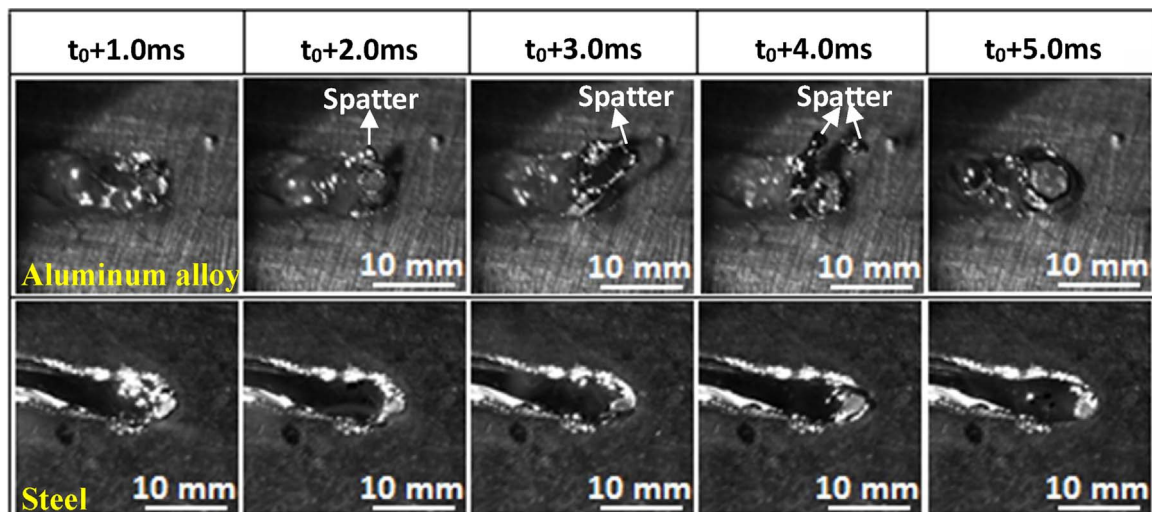


Fig. 8. High-speed images showing behaviors of molten pool and keyhole in laser welding of aluminum alloy and steel.

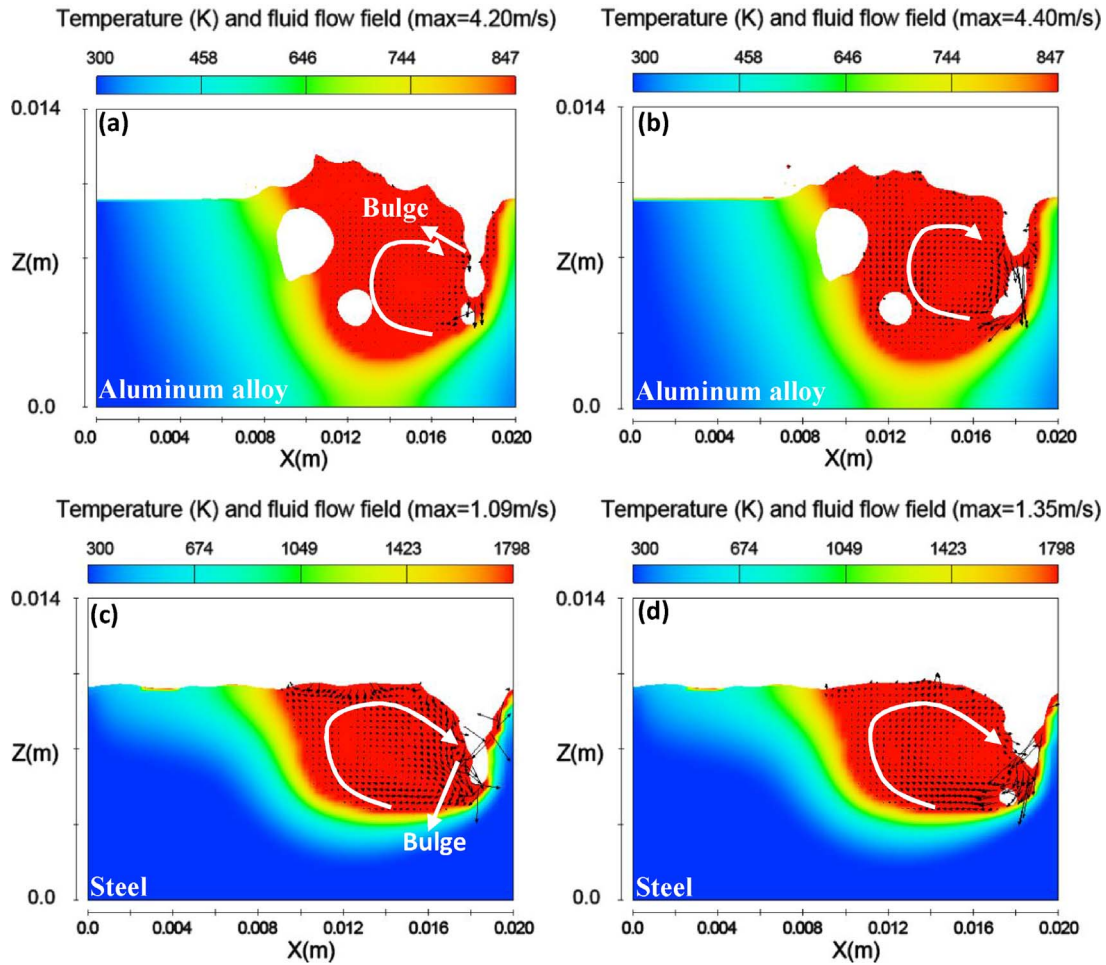


Fig. 9. The process of keyhole collapse during laser welding at 6 kW (a) aluminum alloy at 0.245 s (b) aluminum alloy at 0.250 s (c) steel at 0.264 s (d) steel at 0.270 s.

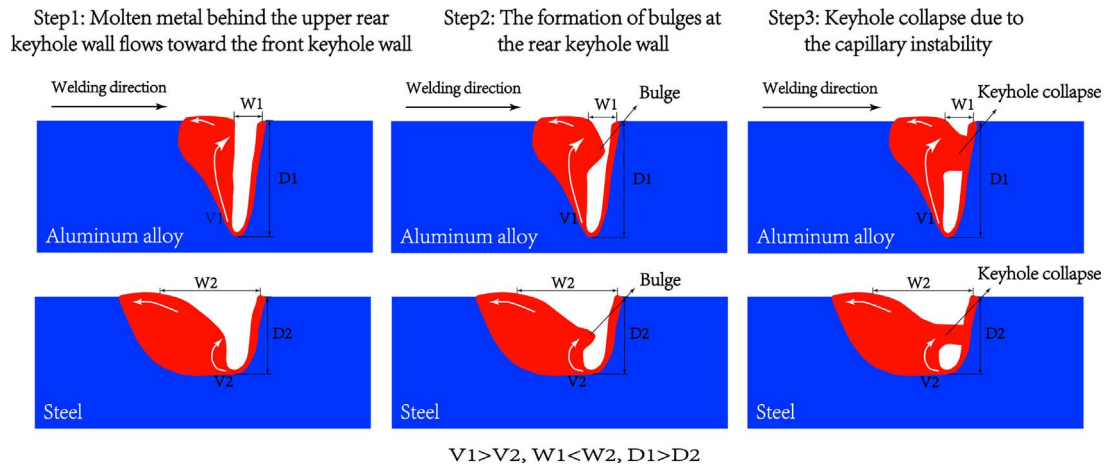


Fig. 10. The diagram of keyhole collapse caused by the keyhole instability.

continue to flow forward and contact with the front wall of the keyhole, causing the keyhole collapse.

- (2) The capillary instability of keyhole. Owing to the much lower melting temperature, higher thermal conductivity of aluminum alloy, deeper weld depth can be obtained during laser welding of aluminum alloy, as shown in Fig. 6. It can also be seen from Fig. 9(a) and (c) that during laser welding of aluminum alloy, the keyhole depth is deeper, and the keyhole entrance size is smaller. When a larger bulge is formed at the upper rear keyhole wall, the

distance between the bulge and front wall of the keyhole is very smaller. Meanwhile, the much larger depth to width ratio of aluminum alloy causes the keyhole capillary instable, which can lead to keyhole collapse between the bulge and front wall of the keyhole, as shown in Fig. 9(b).

In summary, the keyhole collapse caused by the keyhole instability can be described as the following three steps: Molten metal behind the upper rear keyhole wall flows toward the front keyhole wall, the

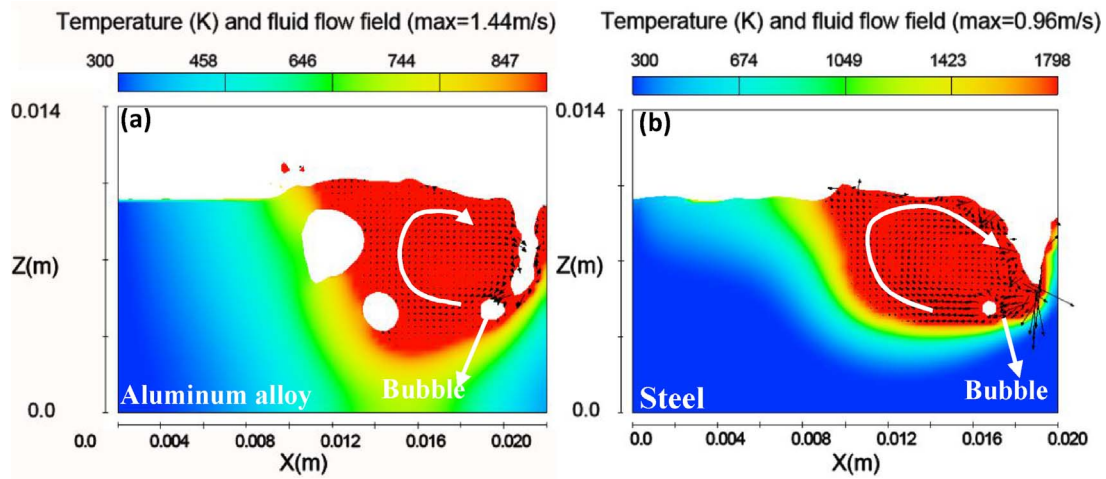


Fig.11. The formation process of bubble at 6 kW (a) laser welding of aluminum alloy at 0.274 s (b) laser welding of steel at 0.280 s.

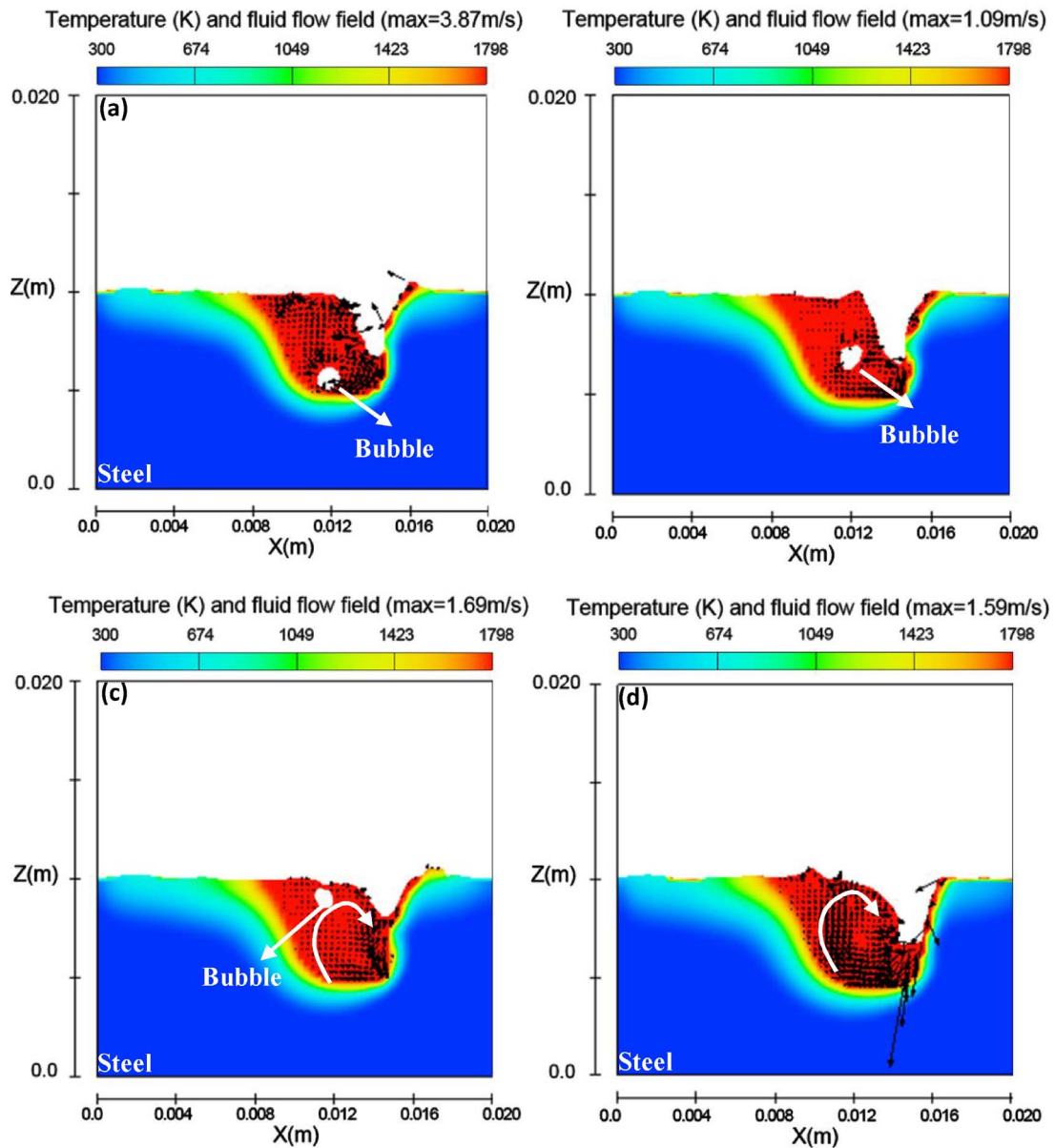


Fig.12. The bubbles escaped from the weld pool in the laser welding of steel (a) $t = 0.129$ s; (b) $t = 0.134$ s; (c) $t = 0.145$ s; (d) $t = 0.150$ s.

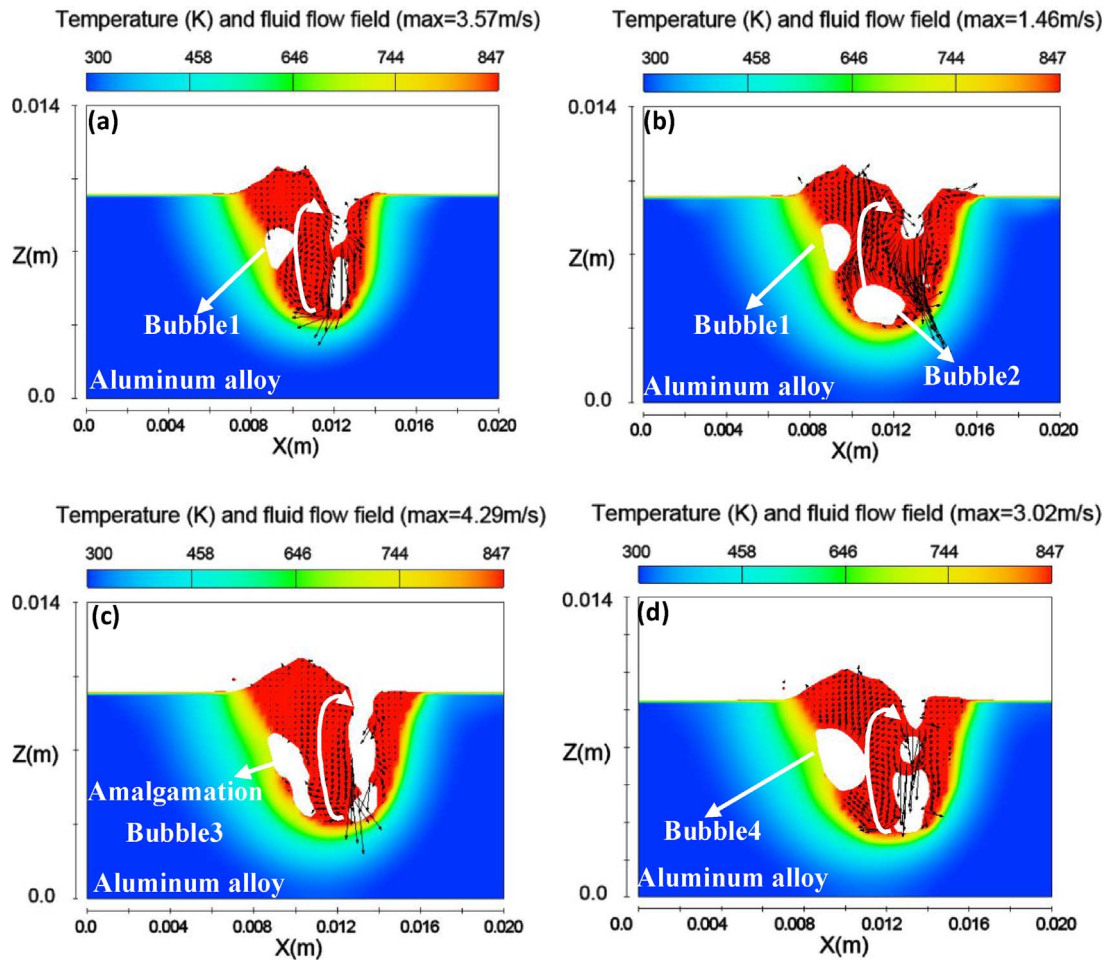


Fig. 13. The bubble migration behavior of movement and transforms into porosity in the laser welding of aluminum alloy (a) $t = 0.080$ s; (b) $t = 0.095$ s; (c) $t = 0.099$ s; (d) $t = 0.105$ s.

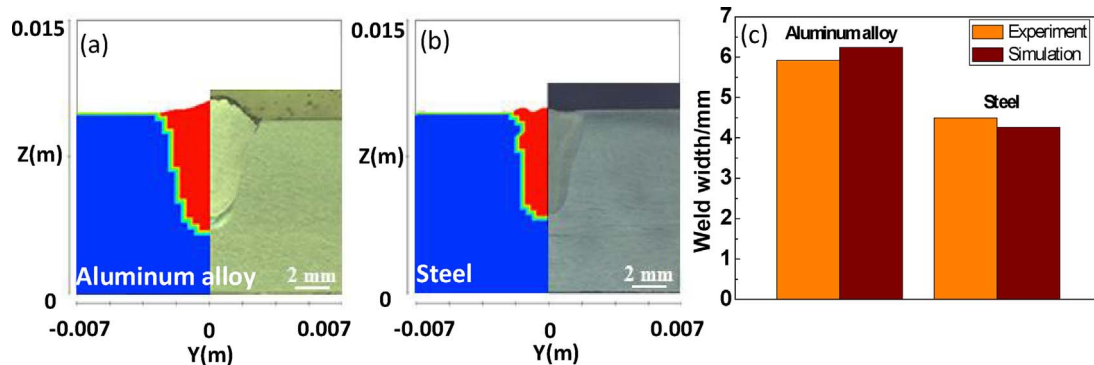


Fig. 14. Comparison of welding profile between numerical and experimental results for (a) aluminum alloy (b) steel, and (c) weld width.

formation of bulges at the rear keyhole wall, and keyhole collapse due to capillary instability. When the keyhole is collapsed, little laser energy can be absorbed by the bottom of keyhole, leading to a shallower keyhole depth. Otherwise, the keyhole depth is deeper. So the frequent keyhole collapse during laser welding of aluminum alloy can easily lead to the large fluctuation of the keyhole depth (Fig. 10).

4.3. Bubble formation and migration

As shown in Fig. 9(b) and (d), when the keyhole collapses, the gas inside the keyhole is trapped, and then a bubble is generated at the bottom of the keyhole. During laser welding of aluminum alloy, the keyhole easily collapses at the middle part of the keyhole, the bubble

size is larger. As shown in Fig. 11(a) and (b), the bubble is pushed backward by the strong clockwise vortex.

After the bubble is generated, it may not change into porosity eventually. Owing to the larger thermal conductivity and higher undercooling degree caused by lower melting point, the solidification rate of aluminum alloy is larger than that of steel. During laser welding of aluminum alloy, the bubble escape from the top surface of weld pool rarely occurs. However, the bubble escape can be readily observed during laser welding of steel, as shown in Fig. 12, at $t = 0.129$ s, bubble forms at the bottom of the keyhole, at $t = 0.134$ s, bubble flows to the middle part of keyhole rear, at $t = 0.145$ s, bubble flows to the top surface of keyhole rear. Due to the low solidification rate of steel, the bubble is far away from the solidification interface, and can not be

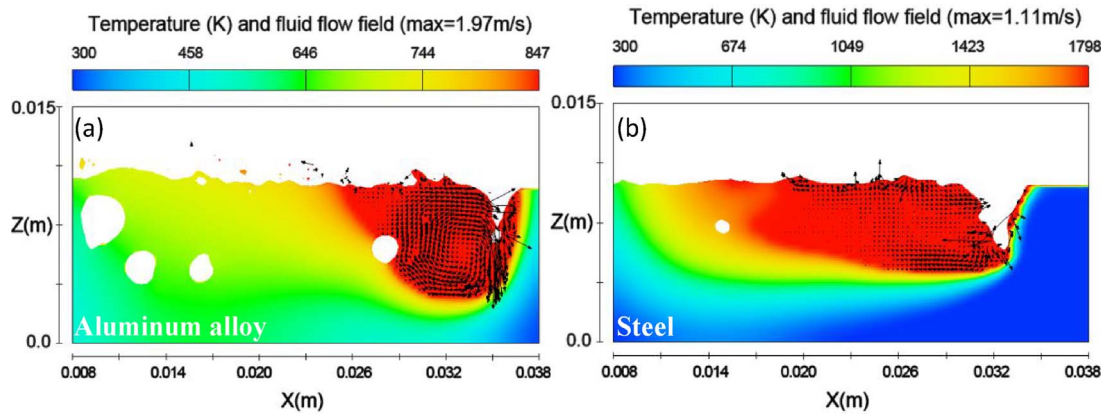


Fig. 15. The porosity number and size in the simulation (weld length 20 mm): (a) aluminum alloy; (b) steel.

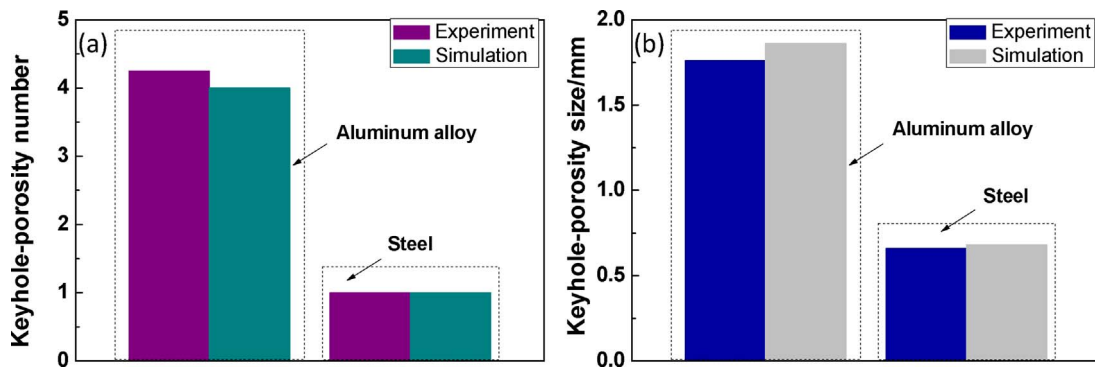


Fig. 16. Comparison of numerical and experimental results (weld length 20 mm): (a) keyhole-porosity number (b) average porosity size.

captured. At $t = 0.150$ s, the bubble escapes from the weld pool.

Fig. 13 shows the bubble capture and bubble amalgamation during laser welding of aluminum alloy. At $t = 0.080$ s, due to the high solidification rate, bubble 1 is captured by the solidification interface; at $t = 0.095$ s, bubble 2 forms at the bottom of the keyhole and flows toward the keyhole rear; at $t = 0.099$ s, bubble 2 combines with bubble 1 to form bubble 3 with larger size; bubble 3 is partially captured; at $t = 0.105$ s, bubble 4 is fully captured by the solidification interface.

During laser welding of steel, owing to little keyhole-induced porosity is formed, bubble amalgamation is difficult to occur.

In summary, the bubble formation caused by the instability of the keyhole, and the bubble captured by the solidification interface, are responsible for the formation of keyhole-induced porosity.

Both melt flow and capillary instability in the weld pool contribute to the stability of the keyhole. During the laser welding of aluminum alloy, the keyhole is more unstable, more bubbles are formed due to the keyhole collapse owing to the more intense melt flow and larger depth to width ratio. Attributed to much larger solidification rate and longer escaping distance, the bubbles are difficultly escape from the weld pool, and more easily captured by the solidification interface. Besides, bubble amalgamation rather than bubble escape is more likely happen. All of these reasons contribute to the formation of keyhole-induced porosity with larger size and number during the laser welding of 5083 aluminum alloy.

4.4. Experiment verification

Due to the violent fluctuation of weld depth in the welding direction, only weld widths in the simulation and experiment are compared. As shown in Fig. 14, the weld width in the simulation and experiment is 6.24 mm and 5.98 mm for aluminum alloy laser welding, and 4.26 mm and 4.50 mm for steel, respectively. The simulative weld geometry sizes are in good agreement with the experimental results.

The porosity number and size are compared with weld length 20 mm. As shown in Fig. 15(a) and (b), in the simulation, the porosity number of aluminum alloy and steel is 4 and 1, respectively. The average porosity size of aluminum alloy and steel is 1.89 mm and 0.68 mm, respectively.

The average porosity size of aluminum alloy and steel based on Fig. 6 is 1.76 mm and 0.66 mm, respectively. The average porosity number and size of experimental and simulative results are shown in Fig. 16(a) and (b). It can be seen that the simulated porosity number and size during laser welding of aluminum alloy and steel agree well with experimental results.

In conclusion, the three-dimensional mathematical model built in this paper could be used legitimately to expound the formation mechanism of keyhole-induced porosity in laser welding of aluminum alloy and steel.

Conclusions

A three-dimensional numerical model is built to study the keyhole instability and the formation mechanism of keyhole-induced porosity in laser welding of aluminum alloy and steel. The major conclusions are listed as follows:

- (1) Although many physical properties between aluminum alloy and steel differ, their laser welding exhibits similar flow patterns inside the weld pools.
- (2) The formation of keyhole-induced porosity can be attributable to the bubble formation caused by the instability of the keyhole, and the bubble captured by the solidification interface.
- (3) In the laser welding of aluminum alloy, owing to the more intense melt flow and larger depth to width ratio, the keyhole is more unstable and prone to collapse, and more bubbles are formed. The keyhole collapse caused by the keyhole instability can be described

as three steps.

- (4) In the laser welding of aluminum alloy, due to much larger solidification rate and longer escaping distance, the bubbles are difficult to escape from the weld pool, and are easily combined. Larger number and size of porosity are formed in laser welding of aluminum alloy.

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